

Smart City Construction, Sustainable Urban Transition, and Carbon Emissions: Nonlinear Evidence from Chinese Cities

Article

Abstract

Smart-city construction increasingly coordinates production, living, ecological, and digital functions, but such coordination does not automatically reduce carbon emissions. This study examines an observed panel of 284 Chinese prefecture-level and above cities from 2006 to 2021. We construct a spatial coupling coordination degree (SCCD) index that treats digital space as a fourth urban functional dimension and estimate two-way fixed-effects models with city and year effects. The full-control observed-sample estimates reveal a significant inverted U-shaped association: SCCD equals 8.444 and SCCD squared equals -8.091, with a turning point of 0.522. Supplementary instrumental-variable evidence implies a turning point of 0.570 but is interpreted cautiously. Industrial upgrading and green technological innovation are associated restructuring pathways rather than confirmed indirect effects. Digital economy development weakens the marginal emission cost of coordination. The results show a governance-sequencing problem: coordination supports low-carbon transition only when cities shift from expansion-oriented integration toward efficiency-oriented operation with stronger data-enabled urban management capacity.

Keywords: smart city construction; spatial coupling coordination; digital space; carbon emissions; governance sequencing; sustainable urban transition

1. Introduction

Cities concentrate population, infrastructure, production, energy use, and emissions, making them a central arena for sustainable transition and carbon mitigation [1-3]. In China, rapid urbanization has unfolded alongside infrastructure expansion, industrial restructuring, and digital transformation, together reshaping how urban functions are connected [4,5]. A central sustainability question is therefore not whether cities coordinate more activities, but whether such coordination reduces frictions faster than it creates construction, reorganization, and energy-demand costs.

Most research treats urban form, compactness, polycentricity, land-use restructuring, and functional zoning as physical spatial problems [6-16]. A related literature treats digitalization as an external driver that may improve information matching, resource optimization, climate adaptation, public-service allocation, and emissions performance [17-25]. These two literatures are rarely integrated. Yet digital infrastructure, platform connectivity, data interaction, and intelligent governance organize flows of information, services, logistics, capital, and ecological monitoring across physical space.

This paper treats digital space as a constitutive urban functional dimension rather than as an external technology variable. It extends the production-living-ecological framework to a production-living-ecological-digital framework and constructs a spatial coupling coordination degree (SCCD) index for 284 Chinese prefecture-level and above cities. The main sample uses observed data from 2006 to 2021. Fitted or extrapolated 2022-2024 observations are excluded from the main evidence and are reserved only for supplementary robustness discussion.

The core finding is a coordination paradox. In the full-control observed-sample specification, SCCD is 8.444*** and SCCD squared is -8.091***. The estimated turning point is 0.522, with a 95% confidence interval of [0.442, 0.601]. The endpoint marginal effects are positive at the low end of the SCCD range and negative at the high end, supporting an inverted U-shaped association rather than a monotonic low-carbon effect.

The distributional implication is important. In the full-control observed baseline sample, 4,375 city-year observations (96.9%) are at or below the turning point, while 139 (3.1%) are above it. This means most observed city-years remain on the expansion-oriented side of coordination, where deeper integration may still carry short-run emission costs.

The paper also examines restructuring pathways and boundary conditions. Industrial upgrading and green technological innovation are treated as associated pathways, not as decomposed indirect effects. Digital economy development is the main supported boundary condition because it attenuates the marginal emission cost of coordination. Environmental regulation and polycentricity are not framed as supported mechanisms unless their generated estimates justify that interpretation.

1.1. Hypotheses

- H1. Spatial coupling coordination across production, living, ecological, and digital spaces has an inverted U-shaped association with log carbon emissions.
- H2. Industrial upgrading and green technological innovation are associated restructuring pathways linking multidimensional coordination to emissions, but they are not interpreted as formal indirect effects in the absence of decomposition estimates.
- H3. Digital economy development attenuates the marginal emission cost of spatial coupling coordination and flattens the nonlinear SCCD-lnCE relationship.
- H4. The nonlinear SCCD-lnCE association differs across major Chinese regions because development stage and governance capacity differ across space.

2. Materials and Methods

2.1. Data and Sample

The main sample covers 284 Chinese prefecture-level and above cities from 2006 to 2021, yielding 4,544 observed city-year observations before regression-specific missing-value restrictions. The full-control baseline uses 4,514 observations after observations with missing controls are removed. The broader working data file contains fitted or extrapolated 2022-2024 observations, but those rows are not treated as observed records in this Sustainability version.

Raw indicators are drawn mainly from the China City Statistical Yearbook, China Energy Statistical Yearbook, China Urban Construction Statistical Yearbook, China Rural Statistical Yearbook, China Industrial Statistical Yearbook, and China Environmental Statistical Yearbook. City-level carbon emissions are matched from EDGAR urban and gridded emission products [30,31].

One data-quality issue is explicitly retained rather than hidden. OPEN contains 72 negative observed-sample values, equal to 1.6% of nonmissing observed OPEN records, with an observed-sample range of [-0.720, 3.640]. Because the supplied analytical data file is the authoritative input, these values are disclosed and retained rather than recoded ex post.

2.2. Variables and SCCD Construction

The dependent variable is log carbon emissions, denoted $\ln CE$. In the available Stata file, the log-transformed outcome is stored under the variable name CE ; the workflow therefore uses CE as $\ln CE$ without applying another logarithmic transformation. The core explanatory variables are $SCCD$ and $SCCD2$. Baseline controls are $OPEN$, UR , URG , and GI .

The $SCCD$ index is built from four functional spaces, 13 criterion layers, and 36 indicators. Positive indicators are normalized as $x^* = (x - \min x) / (\max x - \min x)$, while negative indicators are normalized as $x^* = (\max x - x) / (\max x - \min x)$. Entropy weights are then calculated within functional subsystems. For indicator j , $p_{jit} = x^*_{jit} / \sum(x^*_{jit})$, $e_j = -k \sum(p_{jit} \ln p_{jit})$, $d_j = 1 - e_j$, and $w_j = d_j / \sum(d_j)$.

Subsystem scores are aggregated from the entropy-weighted indicators. The coupling degree C measures the interaction among production, living, ecological, and digital subsystem scores. The comprehensive development index T is the equally weighted average of the four subsystem scores. The final spatial coupling coordination degree is $SCCD = \sqrt{C \times T}$. Equal weights across functional dimensions prevent one subsystem from mechanically dominating the final coordination score.

Table 1. Detailed SCCD indicator system.

Functional space	Criterion layer	Indicator	Calculation	Unit	Dir.
Production	Agricultural production	Gross agricultural output value	Value added of primary industry	10,000 yuan	+
Production	Agricultural production	Per capita crop sown area	Sown area / population	ha / 10,000 persons	+
Production	Agricultural production	Per capita aquatic product output	Aquatic product output / population	tons / 10,000 persons	+
Production	Agricultural production	Per capita grain output	Grain output / population	tons / person	+
Production	Industrial production	Secondary industry output value	Secondary industry output value	10,000 yuan	+
Production	Industrial production	Total profits of industrial enterprises above designated size	Total profits of industrial enterprises above designated size	10,000 yuan	+
Production	Industrial production	Number of manufacturing employees	Number of manufacturing employees	10,000 persons	+
Production	Other production functions	Tertiary industry output value	Tertiary industry output value	10,000 yuan	+
Production	Other production functions	Postal business revenue	Postal business revenue	10,000 yuan	+
Production	Other production functions	Total retail sales of consumer goods	Total retail sales of consumer goods	10,000 yuan	+
Production	Other production functions	Total imports and exports of goods	Total imports and exports of goods	10,000 yuan	+
Living	Basic living	Population density	Year-end population / urban area	10,000 persons / km ²	+

Living	Basic living	Road area per capita	Road area per capita	m2 / person	+
Living	Basic living	Completed investment in residential development	Completed investment in residential development	10,000 yuan	+
Living	Basic living	Average wage of employed staff	Average wage of employed staff	yuan / person	+
Living	Basic living	Per capita general public budget expenditure	Per capita general public budget expenditure	yuan / person	+
Living	Medical living	Hospital and health center beds per 10,000 persons	Hospital beds / population	beds / 10,000 persons	+
Living	Medical living	Licensed physicians per 10,000 persons	Licensed physicians / population	persons / 10,000 persons	+
Living	Educational living	Per capita education expenditure	Education expenditure / population	yuan / person	+
Living	Educational living	Teachers per 10,000 persons	Primary + secondary + university teachers / population	persons / 10,000 persons	+
Living	Science and technology living	Per capita science and technology expenditure	Science and technology expenditure / population	yuan / person	+
Living	Science and technology living	Invention patents granted per 10,000 persons	Number of invention patents granted / population	units / 10,000 persons	+
Living	Science and technology living	Employees in scientific research and integrated technical services	Employees in scientific research and integrated technical services	10,000 persons	+
Living	Cultural living	Library collection per 10,000 persons	Library collection / urban population	volumes / 10,000 persons	+
Ecological	Environmental pressure	Industrial wastewater discharge per unit area	Industrial wastewater discharge / urban area	10,000 tons / km2	-
Ecological	Environmental pressure	Industrial sulfur dioxide emissions per unit area	Industrial sulfur dioxide emissions / urban area	10,000 tons / km2	-
Ecological	Environmental pressure	Industrial smoke and dust emissions per unit area	Industrial smoke and dust emissions / urban area	10,000 tons / km2	-
Ecological	Environmental response	Employees in water conservancy, environment and public facilities management	Employees in water conservancy, environment and public facilities management	10,000 persons	+
Ecological	Environmental status	Green coverage rate of built-up area	Green coverage rate of built-up area	%	+
Ecological	Environmental status	Green space per capita	Urban green space area / population	m2 / person	+
Digital	Informatization	Internet broadband access users	Internet broadband access users	10,000 persons	+
Digital	Informatization	Telecommunication service volume per 10,000 persons	Total telecommunication services / population	units / 10,000	+

				persons	
Digital	Digital-intelligent	Digital and smart enterprises per 10,000 persons	Digital firms + smart enterprises	enterprises / 10,000 persons	+
Digital	Networked	IoT enterprises per 10,000 persons	Number of IoT enterprises identified from Qichacha / population	enterprises / 10,000 persons	+
Digital	Networked	Industrial internet enterprises per 10,000 persons	Number of industrial internet enterprises identified from Qichacha / population	enterprises / 10,000 persons	+
Digital	Networked	Satellite internet enterprises per 10,000 persons	Number of satellite internet enterprises identified from Qichacha / population	enterprises / 10,000 persons	+

Note: The indicator system follows the production-living-ecological-digital framework after deleting one policy-term frequency indicator. The Stata workflow uses the generated composite SCCD and does not recalculate raw entropy weights.

2.3. Empirical Strategy

The baseline specification is $CE_{it} = \beta_1 SCCD_{it} + \beta_2 SCCD2_{it} + \gamma X_{it} + city_i + year_t + \epsilon_{it}$, where X_{it} contains OPEN, UR, URG, and GI. City fixed effects absorb time-invariant city characteristics, year fixed effects absorb common annual shocks, and standard errors are clustered by city.

The inverted-U interpretation requires $\beta_1 > 0$, $\beta_2 < 0$, a turning point inside the observed SCCD range, and endpoint marginal effects with the expected signs. The workflow therefore reports the turning point, its standard error and confidence interval, marginal effects at the SCCD minimum, mean, and maximum, and the observed-sample distribution relative to the turning point.

Robustness checks include 1%-99% winsorized variables, night-time light activity as an alternative outcome check, and MCCD as an alternative coordination proxy when available. Supplementary endogeneity analysis instruments SCCD and SCCD2 with the pre-existing lagged SCCD instrument IV and IV2. The workflow does not recalculate or overwrite IV or c_SCCD.

Mechanism-equation evidence examines industrial upgrading (OIU) and green technological innovation (GTI). These equations are interpreted as associated restructuring evidence rather than formal mediation or decomposed indirect effects. Moderation models examine DEI, ER, and POLY, but the text focuses on DEI because it is the supported boundary condition in the generated evidence.

3. Results

3.1. Descriptive Statistics

Table 2 summarizes the observed 2006-2021 sample. The observed sample contains no fitted rows, and regression-specific samples vary only because of missing controls or instruments.

Table 2. Descriptive statistics for the observed 2006-2021 sample.

Variable	N	Mean	S.D.	Min	Max
CE	4544	4.528	1.292	0.000	8.325
SCCD	4544	0.313	0.086	0.121	0.822
SCCD2	4544	0.106	0.068	0.015	0.676

OPEN	4529	0.193	0.345	-0.720	3.640
UR	4544	53.912	15.862	6.491	100.000
URG	4529	2.466	0.567	1.207	6.378
GI	4544	0.185	0.101	0.043	1.027
DEI	4544	0.056	0.077	0.000	0.940
POLY	4544	0.347	0.194	0.000	0.963
OIU	4544	2.278	0.149	1.163	2.836
GTI	4544	3.931	1.863	0.000	9.872
IV	4260	0.317	0.087	0.121	0.822

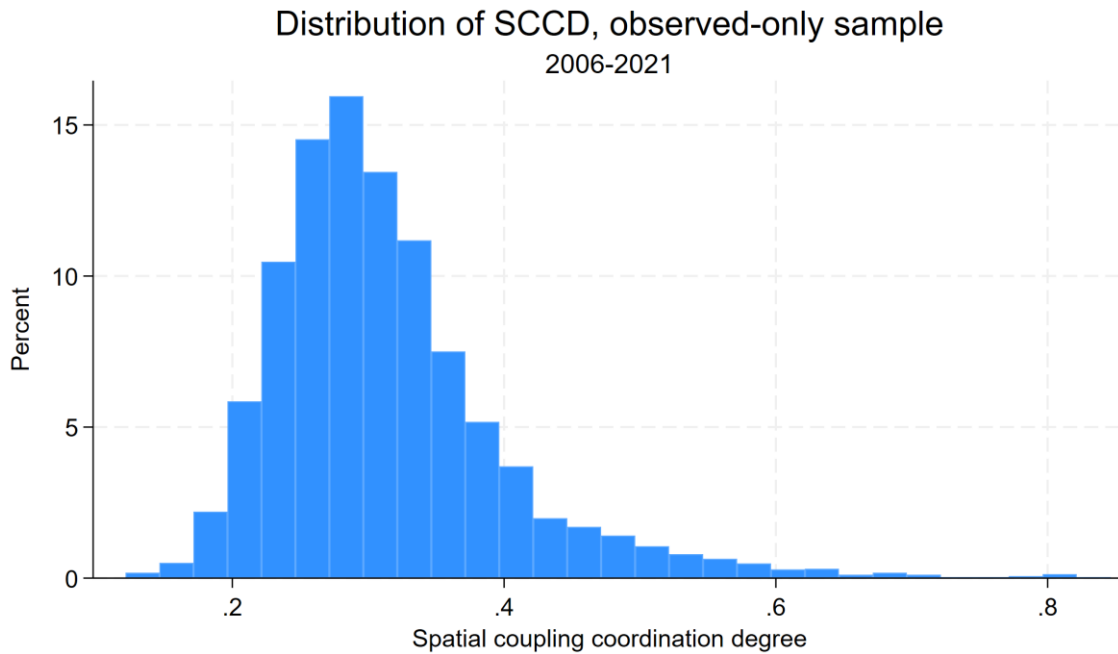


Figure 1. Distribution of SCCD in the observed 2006-2021 sample.

3.2. Baseline Nonlinear Relationship

Table 3 reports the baseline two-way fixed-effects estimates. The full-control observed-sample specification gives $SCCD = 8.444^{***}$ and $SCCD2 = -8.091^{***}$. The implied turning point is 0.521838, and the confidence interval remains inside the observed SCCD range. The marginal effect is 6.479 at the left endpoint, 3.386 at the sample mean, and -4.861 at the right endpoint. This supports H1 and rejects a simple monotonic sustainability story.

Table 3. Baseline regression results: observed-only sample, 2006-2021.

	(1)	(2)	(3)	(4)	(5)
CE	CE	CE	CE	CE	
SCCD	12.957***	11.841***	9.007***	8.819***	8.444***

	(1.445)	(1.421)	(1.305)	(1.348)	(1.338)
Spatial coupling coordination degree squared	- 11.476***	-10.296***	-8.400***	-8.125***	-8.091***
	(1.749)	(1.761)	(1.593)	(1.642)	(1.665)
OPEN		0.456***	0.211**	0.244***	0.276***
	(0.108)	(0.083)	(0.083)	(0.082)	
UR			0.029***	0.027***	0.025***
	(0.004)	(0.004)	(0.004)		
URG				-0.116*	-0.101
	(0.068)	(0.067)			
GI					-1.953***
	(0.546)				
Constant	1.203***	1.314***	0.629**	1.053***	1.497***
	(0.284)	(0.272)	(0.267)	(0.357)	(0.382)
N	4544	4529	4529	4514	4514
Within R- squared	0.592	0.601	0.640	0.641	0.650

Notes: City and year fixed effects are included. Robust standard errors clustered by city are in parentheses. *, **, and *** denote significance at 10%, 5%, and 1%, respectively.

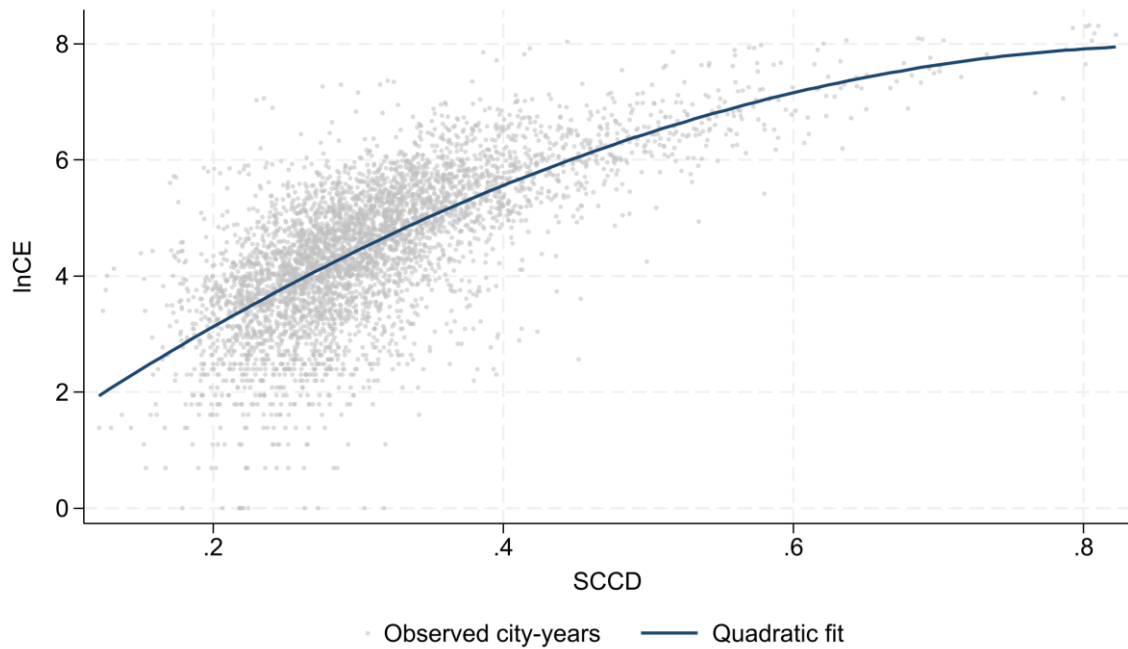


Figure 2. Observed SCCD-InCE relationship and quadratic fit, 2006-2021.

3.3. Robustness and Supplementary IV Evidence

Table 4 shows that the nonlinear pattern remains under winsorization, the night-time light activity check, and supplementary IV estimation. The observed-only IV turning point is 0.570248. The first-stage diagnostics report F statistics of 61.55 for SCCD and 80.44 for SCCD2; the Kleibergen-Paap rk Wald F statistic is n.a.. These diagnostics are reported to make the IV evidence auditable, but the IV estimates remain supplementary because lagged SCCD cannot eliminate all identification concerns in an observational panel.

Table 4. Robustness and supplementary IV results: observed-only sample, 2006-2021.

	(1)	(2)	(3)	(4)
w_CE	NTL	CE	CE	
w_SCCD	9.845***			
	(1.337)			
w_SCCD2	-10.134***			
	(1.648)			
w_OPEN	0.385***			
	(0.117)			
w_UR	0.025***			
	(0.004)			
w_URG	-0.054			
	(0.068)			
w_GI	-1.968***			
	(0.548)			
SCCD		3.601***		17.971***
	(0.719)		(2.816)	
SCCD2		-2.765***		-15.757***
	(0.763)		(2.969)	
OPEN		0.264***	0.276***	0.176**
	(0.090)	(0.082)	(0.080)	
UR		0.019***	0.025***	0.018***
	(0.003)	(0.004)	(0.004)	

URG		-0.055	-0.101	-0.078
	(0.062)	(0.067)	(0.065)	
GI		-1.668***	-1.953***	-1.587***
	(0.362)	(0.546)	(0.564)	
MCCD			8.444***	
	(1.338)			
MCCD2			-8.091***	
	(1.665)			
Constant	1.122***	-0.344	1.497***	-0.179
	(0.386)	(0.293)	(0.382)	(0.549)
N	4514	4514	4514	4232
Within R-squared	0.658	0.583	0.650	0.626

Notes: IV estimates use the pre-existing lagged SCCD instrument and its square. First-stage and weak-instrument diagnostics are reported in the generated supplementary diagnostic files.

3.4. Regional Heterogeneity

Table 5 indicates that the inverted U-shaped pattern appears across major regions, but coefficient magnitudes differ. This supports H4 and implies that the same increase in coordination may have different emission implications across regional development stages.

Table 5. Regional heterogeneity: observed-only sample, 2006-2021.

	(1)	(2)	(3)
Eastern China	Central China	Western China	
SCCD	7.368***	12.447***	7.842***
	(2.557)	(2.490)	(1.740)
SCCD2	-6.905***	-12.828***	-8.152***
	(2.543)	(3.179)	(2.102)
OPEN	0.225***	0.416*	0.258
	(0.085)	(0.251)	(0.305)
UR	0.027***	0.022***	0.024***
	(0.006)	(0.006)	(0.007)
URG	-0.069	-0.018	-0.187*
	(0.143)	(0.121)	(0.103)

GI	-1.098	-2.292***	-1.871**
	(1.134)	(0.667)	(0.885)
Constant	1.615**	0.545	1.920***
	(0.711)	(0.700)	(0.639)
N	1537	1585	1392
Within R-squared	0.731	0.645	0.624

3.5. Associated Mechanism Equations

Table 6 reports associated mechanism equations for OIU and GTI. SCCD is positively associated with both industrial upgrading and green technological innovation, while the squared terms are negative. Once OIU or GTI enters the CE equation, the nonlinear SCCD pattern remains. The evidence is therefore consistent with H2 as restructuring association, not as a quantified pathway effect.

Table 6. Associated mechanism equations: observed-only sample, 2006-2021.

	(1)	(2)	(3)	(4)
OIU	CE with OIU	GTI	CE with GTI	
SCCD	0.647***	7.561***	10.379***	5.290***
	(0.160)	(1.254)	(1.402)	(1.115)
SCCD2	-0.407*	-7.535***	-5.775***	-6.336***
	(0.207)	(1.503)	(1.468)	(1.418)
OIU		1.366***		
	(0.316)			
GTI				0.304***
	(0.027)			
OPEN	0.008	0.265***	0.224***	0.208***
	(0.012)	(0.077)	(0.085)	(0.071)
UR	0.004***	0.020***	0.021***	0.019***
	(0.000)	(0.004)	(0.004)	(0.003)
URG	0.029***	-0.141**	0.028	-0.109*
	(0.006)	(0.066)	(0.064)	(0.063)
GI	-0.093*	-1.826***	-2.102***	-1.314**
	(0.049)	(0.521)	(0.403)	(0.552)
Constant	1.808***	-0.972	-0.725*	1.718***

	(0.041)	(0.697)	(0.411)	(0.325)
N	4514	4514	4514	4514
Within R-squared	0.622	0.658	0.826	0.686

Notes: These are pathway equations rather than formal indirect-effect estimates.

3.6. Moderation and Boundary Conditions

Table 7 reports observed-sample moderation estimates. DEI significantly attenuates the nonlinear SCCD-InCE relationship, supporting H3. Conditional calculations show a marginal effect at the SCCD mean of 3.792 at low DEI and 3.094 at high DEI. POLY is retained only as a supplementary interaction check and is not treated as a supported moderation mechanism.

Table 7. Moderation analysis: observed-only sample, 2006-2021.

	(1)	(2)	(3)
ER	DEI	POLY	
SCCD	10.931***	8.869***	7.925***
	(1.991)	(1.198)	(2.146)
ER	61.643		
	(46.315)		
SCCD # ER	-325.788		
	(228.112)		
SCCD2	-11.675***	-7.977***	-8.678***
	(2.435)	(1.423)	(2.660)
ER	0.000		
	(.)		
SCCD2 # ER	455.053*		
	(266.493)		
OPEN	0.278***	0.263***	0.277***
	(0.082)	(0.083)	(0.082)
UR	0.025***	0.025***	0.024***
	(0.004)	(0.004)	(0.004)
URG	-0.097	-0.090	-0.090
	(0.067)	(0.067)	(0.067)
GI	-1.962***	-2.008***	-1.954***

	(0.536)	(0.544)	(0.532)
DEI		5.874***	
	(2.226)		
SCCD # DEI		-25.913***	
	(7.936)		
DEI		0.000	
	(.)		
SCCD2 # DEI		24.173***	
	(6.665)		
POLY			-1.043
	(0.941)		
SCCD # POLY			2.842
	(4.782)		
POLY			0.000
	(.)		
SCCD2 # POLY			0.159
	(5.485)		
Constant	1.040**	1.364***	1.763***
	(0.486)	(0.370)	(0.487)
N	4514	4514	4514
Within R-squared	0.651	0.652	0.653

Notes: ER, DEI, and POLY are estimated in separate interaction models. The text focuses on DEI as the supported boundary condition.

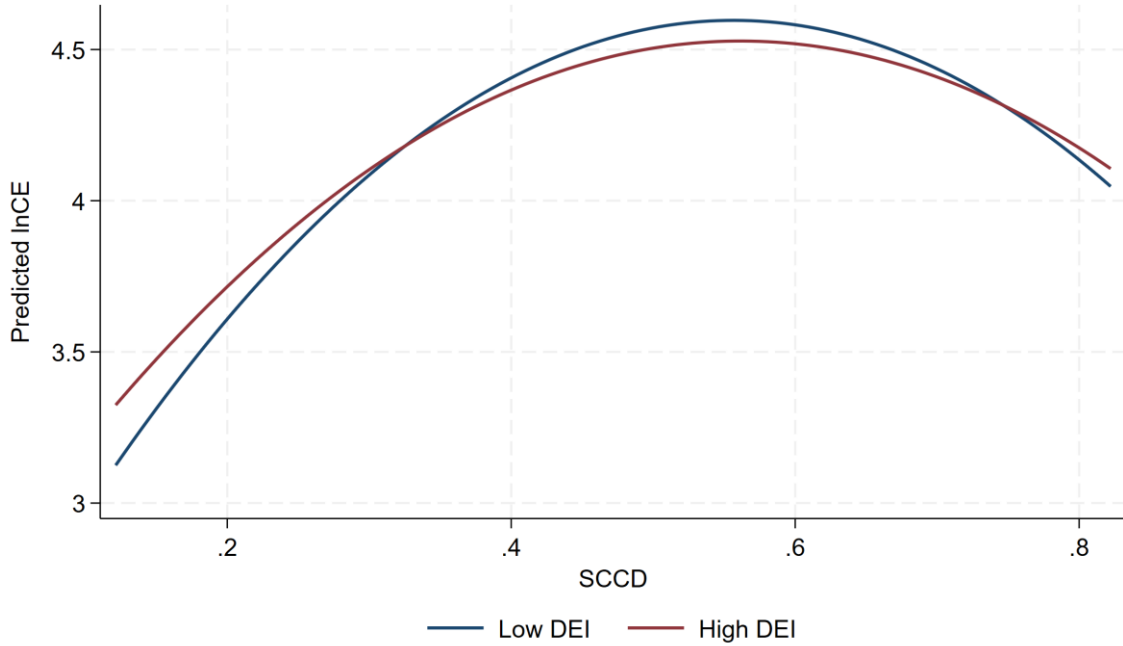


Figure 3. Predicted SCCD-InCE curves at low and high DEI levels, observed sample.

4. Discussion

The results clarify why smart-city construction and spatial coordination do not automatically reduce emissions. At low coordination levels, cities often connect production, living, ecological, and digital functions through infrastructure expansion, industrial concentration, construction activity, and additional energy input. These expansion effects can dominate early efficiency gains and generate the upward side of the inverted U-shaped relationship.

At higher coordination levels, the same process can become more efficiency-oriented. Better information matching, integrated public services, smart transport, ecological monitoring, and cross-departmental governance can reduce redundant investment and improve resource allocation. The turning point should therefore be read as a governance-stage marker, not as a universal policy target.

The mechanism-equation results explain why the transition is gradual. Industrial upgrading and green technological innovation may support long-run low-carbon transition, but they also require new investment, equipment replacement, experimentation, and commercialization. These processes can create short-run carbon pressure before efficiency gains materialize.

The DEI results are the clearest Sustainability-facing boundary condition. Digital economy development appears to weaken the marginal emission cost of coordination, suggesting that digital capacity matters when it becomes operational capability: interoperable data systems, energy-management platforms, smart mobility tools, and governance routines that convert spatial coordination into efficiency rather than duplicated construction.

Several limitations remain. The SCCD index captures key aspects of digital space, but it cannot fully observe platform governance, data mobility, algorithmic coordination, or digital public-service quality. The empirical design includes fixed effects, robustness checks, and supplementary IV estimation, but the

analysis remains observational. The 2022-2024 fitted or extrapolated rows are excluded from the main evidence to avoid overstating data coverage.

5. Conclusions

Using an observed panel of 284 Chinese cities from 2006 to 2021, this study finds an inverted U-shaped association between SCCD and log carbon emissions. The full-control observed-sample turning point is 0.521838, and supplementary IV estimates imply a turning point of 0.570248. Coordination is therefore not inherently low-carbon; its emission implication depends on whether governance remains expansion-oriented or becomes efficiency-oriented.

The policy implication is governance sequencing. Cities below the turning point should avoid converting smart-city construction into duplicated infrastructure, land expansion, and energy-intensive restructuring. Cities approaching the turning point should strengthen data-enabled governance, energy management, integrated public services, and coordination between industrial and innovation policy. Regional strategies should differ because eastern, central, and western cities display different responsiveness to SCCD.

Future research should build richer measures of digital urban governance, examine longer dynamic lags in industrial upgrading and green innovation, and test whether the SCCD framework travels to other national and metropolitan contexts. More direct causal designs would also help distinguish coordination effects from unobserved policy and development shocks.

Author Contributions

The working files used to generate this analytical manuscript did not contain final CRediT roles. The author team must finalize this statement against the named author list before journal submission.

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Institutional Review Board Statement

Not applicable. This study uses city-level statistical and emissions data and does not involve human participants or animal subjects.

Informed Consent Statement

Not applicable.

Data Availability Statement

The Stata do-files, Python generation script, generated tables, and generated figures can be made available in a public replication repository. The raw and processed city-level analytical data draw on

statistical yearbooks, EDGAR products, and third-party enterprise-query data; redistribution may be limited by source licensing. Data can therefore be provided by the authors upon reasonable request where source licenses permit.

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No acknowledgement text was available in the working files used to generate this analytical manuscript.

Conflicts of Interest

Conflict-of-interest metadata were not available in the working files used to generate this analytical manuscript. The final submission must include the author team's formal conflict-of-interest declaration.

Supplementary Materials

The supplementary evidence generated for this revision includes the observed-sample U-test output, IV diagnostics, DEI conditional turning-point calculations, OPEN data-quality diagnostic, full 2006-2024 fitted/extrapolated sensitivity tables, and the code used to generate tables and figures. The 2022-2024 fitted or extrapolated observations are supplementary sensitivity evidence and are not observed statistical records.

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